V - Vector space

↳ how do we create $W < V$

↑ Subspace

[you have to be creative] ← practice

Create W:

↳ pick out of any finite # of elements

$$\{V_1, V_2 \dots V_n\}$$

-> Why?

→ Suppose we have a homgenious system of L.E.

$$A \cdot \bar{X} = 0$$

$\dim A = m \times n$

$\dim(\bar{X}) = n \times 1$

$$\bar{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Clearly any solution $\in \mathbb{R}^n$

Theorem Solution space of hom S.L.E is a subspace of $\mathbb{R}^n$

↓ Proof

$A \cdot \bar{X} = \bar{0}$

$A \cdot \bar{Y} = \bar{0}$

$\bar{X}$ and $\bar{Y}$ are solutions

$W_1 \; W_2 \in W$

① $W_1 + W_2 \in K$

② $K \cdot W_1 \in W$

↳ – $\bar{X} + \bar{y}$ – (?) is it a solution?

↳ $A \cdot [\bar{X} + \bar{Y}] = A \cdot \bar{X} + A \cdot \bar{Y} = \bar{0} + \bar{0} = 0$

$\therefore \bar{X} + \bar{Y}$ is a solution

– multiplication by K

$A[K \cdot \bar{X}] = K[\quad]$

matrix multiply ↑ $\bar{0}$

=> proof is complete

② $A \cdot \bar{x} = \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$

find all solutions of above $\Rightarrow$ is it a subspace of $\mathbb{R}^n$

↳ any subspace must have 0 element $\rightarrow \bar{x} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$

$$A \cdot \bar{x} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \neq \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$$

$\Rightarrow$ it's only a subspace if S.L.E is homogenous

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

So we know

W is Subspace

Solve this ↗

$x - 2y + 3z = 0$ ↩ $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Plane in $\mathbb{R}^3$, $P(0,0,0)$

$n = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

Liniar independence

V is a vector space

always given -> $\{\vec{v}_1, \vec{v}_2, \dots \vec{v}_n\} \in V$

are they liniar independent?

S.L.E

if $\boxed{c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = \vec{0}}$

it has multiple solutions if $c_1 = c_2 = c_3 = \dots = c_n = 0$

Ex $V = \mathbb{R}^2$

↳ $V_1 \dots V_n$ → obviously $n \geq 2$

$n = 2$

$$V_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad V_2 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

↳ $c_1\vec{V}_1 + c_2\vec{V}_2 = \vec{0}$

unknown from linear combination

↳ $\begin{bmatrix} \vec{V}_1 & \vec{V}_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

↳ $\begin{bmatrix} 1 & 6 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

↳ solve

tricky part

1 - solution → YAY

∞ - solution → NAY

$V_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $V_2 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$ $V_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 6 & 1 \\ 2 & 7 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

↳ use augmented matrix approach

↳ $\begin{bmatrix} 1 & 6 & 1 & 0 \\ 2 & 7 & 0 & 0 \end{bmatrix}$ REF →

↓

blah

blah

blah

if $\det(A) = 0$ → ∞

$\det(A) \neq 0$ → 1 solution

$$V_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad V_2 = \begin{bmatrix} -2 \\ 2 \\ 4 \end{bmatrix} \quad V_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$C_1 + \quad + C_2 + \quad + C_3 + \quad = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 2 \\ 2 & 2 & 0 \\ 1 & 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 2 & 2 & 0 \\ 1 & 4 & 1 \end{bmatrix} \rightarrow \det(A)$$

$$\det(A) = \underbrace{2}_{a_{21}} \cdot \underbrace{1}_{-M_{21}} \cdot C_{21} + \underbrace{2}_{a_{22}} \cdot \underbrace{C_{22}}_{M_{22}} + \cancel{\underbrace{0}_{a_{23}} \cdot C_{23}}$$

$$= -2M_{21} + 2M_{22}$$

$$M_{21} - \begin{bmatrix} -2 & 2 \\ 4 & 1 \end{bmatrix} = \det\begin{bmatrix} -2 & 2 \\ 4 & 1 \end{bmatrix} = -2 - 8 = -10$$

$$M_{22}\begin{bmatrix} 1 & -1 & -2 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix} = \det\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} = 1 - 2 = -1$$

Any set of vectors that contain _____

$\Rightarrow \{V_1, V_2, 0\}$

↑ Zero

$C_1 V_1 + C_2 V_2 + C_3 \cdot 0 = 0$

$\begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix}$ -?

$C_1 = 0$
$C_2 = 0$
$C_3 = t$

infinit number of solutions for t

$0 \cdot V_1 + 0 \cdot V_2 + t \cdot 0 = 0$

↑ any value

det(A) does not work for

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 0 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ -2 & 3 & 1 & 0 \end{bmatrix}$$ ← Not Square

RREF →

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

↑ No leading 1

Linear dependent

∴ free variable and ∞ solutions

↓

of vectors < n

of $\mathbb{R}^n$

$n=4$

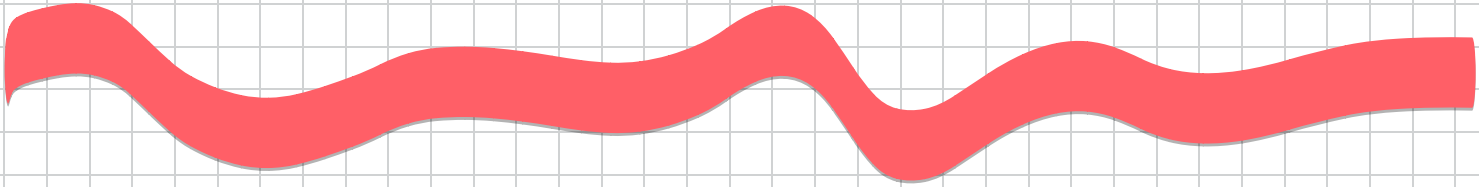
↑ 3

When you have to guess linear dependance

$\dim(\mathbb{R}^n) = n$

$S = \{\vec{v}_1, \vec{v}_2, \ldots \vec{v}_r\}$ and $r > n$

S is always linear dependent



you can sensibly ask about lin dep/indep only if you have 2 or more vectors

if two $\{v_1, v_2\}$ linear dependent

$$c \cdot v_1 + c_2 v_2 = \vec{0}$$

if linear dependent

$$c_1^2 + c_2^2 \neq 0$$

assume $c_1 \neq 0 \Rightarrow \vec{v_1} = -\frac{c_2}{c_1}\vec{v_2}$

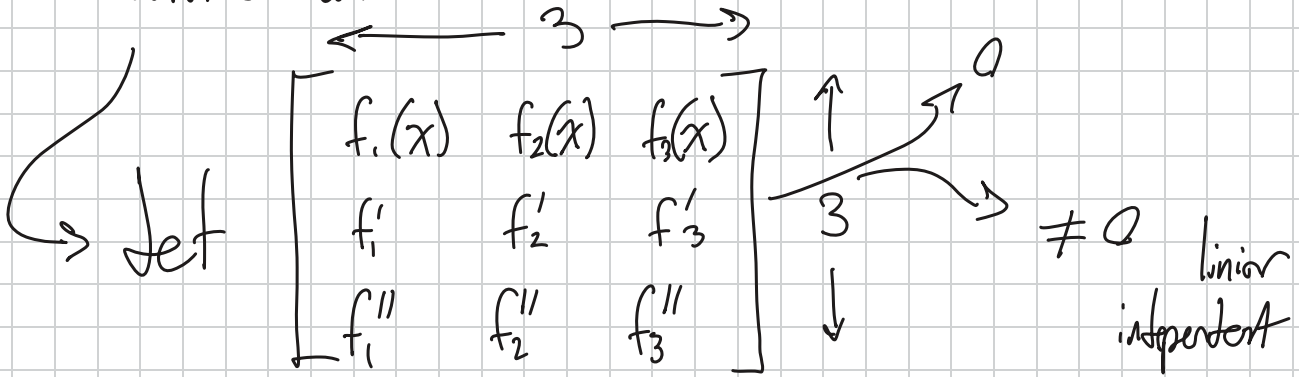
$\Rightarrow \vec{f}$ - real vector space of functions

$$f_1(x), \quad f_2(x), \quad f_3(x)$$

Why functions are linearly dependent

Theorem

↳ "Wronskian"



Ex $f_1 = \sin x$

$f_2 = \cos x$

$$\det \begin{bmatrix} \sin x & \cos x \\ (\sin x)' & (\cos x)' \end{bmatrix} = \det \begin{bmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{bmatrix} =$$

$$= \sin x \cdot (-\sin x) - \cos x \cos x =$$

$$= -\sin^2 x - \cos^2 x$$

$$= -1$$

Never "0"
no matter the "x"

$$P_4 = \text{Span}\{1, x, x^2, x^3, x^4\}$$

↑ vector space of polynomials of degree ≤ 4

↑ → any $\deg(P) \leq 4$ can be built from the spanning set

Are they linearly indep?

$$\begin{bmatrix} 1 & x & x^2 & x^3 & x^4 \\ 0 & 1 & 2x & 3x^2 & 4x^3 \\ 0 & 0 & 2 & 6x & 12x^2 \\ 0 & 0 & 0 & 6 & 24x \\ 0 & 0 & 0 & 0 & 24 \end{bmatrix}$$